Fiscal Year:	FY 2004	**Task Last Updated:**	FY 01/05/2005
PI Name:	McNeil, Paul L Ph.D.		
Project Title:	Cellular and Molecular Mechansims Mediating Rapid Resealing of Gravity-induced Plasma Membrane Disruption		
Division Name:	Space Biology		
Program/Discipline:	SPACE BIOLOGY		
Program/Discipline--Element/Subdiscipline:	SPACE BIOLOGY--Cellular and molecular biology		
Joint Agency Name:		**TechPort:**	No
Human Research Program Elements:	None		
Human Research Program Risks:	None		
Space Biology Element:	None		
Space Biology Cross-Element Discipline:	None		
Space Biology Special Category:	None		
PI Email:	pmcneil@mail.mcg.edu	**Fax:**	FY
PI Organization Type:	UNIVERSITY	**Phone:**	(706) 721-3065
Organization Name:	Medical College of Georgia		
PI Address 1:	CA 2018		
PI Address 2:			
PI Web Page:			
City:	Athens	**State:**	GA
Zip Code:	30912	**Congressional District:**	12
Comments:			
Project Type:	Ground	**Solicitation / Funding Source:**	99-HEDS-02
Start Date:	07/01/2000	**End Date:**	06/30/2004
No. of Post Docs:	2	**No. of PhD Degrees:**	0
No. of PhD Candidates:	0	**No. of Master' Degrees:**	0
No. of Master's Candidates:	0	**No. of Bachelor's Degrees:**	0
No. of Bachelor's Candidates:	0	**Monitoring Center:**	NASA ARC
Contact Monitor:		**Contact Phone:**	
Contact Email:			
Flight Program:			
Flight Assignment:			
Key Personnel Changes/Previous PI:	4		
COI Name (Institution):			
Grant/Contract No.:	NAG2-1416		
Performance Goal No.:			
Performance Goal Text:			
Task Description:	We are attempting to elucidate the molecular mechanism used by skeletal muscle myocytes in repairing tears in their surface membrane. This requires the development of novel microsope-based technologies that will allow us to make membrane disruptions in living myocytes and then assess resealing efficiency, and it requires the identification and/or generation of relevant genetically engineered mouse models. This study has profound implications for our understanding normal muscle phsyiology, and hence pathologies, such as those encountered during space flight.		
Rationale for HRP Directed Research:			
Research Impact/Earth Benefits:	This study represents the first report of a disease that results from a resealing deffect.		

Task Progress:	Disruption or tearing of the plasma membrane of skeletal muscle myocytes is a normal occurrence in mechanically loaded skeletal muscle. In order to survive this injury the myocyte must be capable of rapid repair. One cardinal prediction of these hypotheses is that disease will result from a defect in the myocyte capacity for repair. The primary task of this period was to test this hypothesis. We developed a novel, laser-based method for reproducibly creating disruptions in myocyte membranes, and quantitating resealing efficacy. This technology was used to measure resealing in mice genetically engineered to be defective in production of the protein dysferlin, which other studies had suggested might play role in repair. We found that the myocytes from this mutant mouse were severely defective in resealing.
Bibliography Type:	Description: (Last Updated: 06/11/2008)
Articles in Peer-reviewed Journals	McNeil, P.L., K. Miyake and S.S. Vogel "The endomembrane requirement for cell surface repair" PNAS 100: 4592-4597 , Apr-2003
Articles in Peer-reviewed Journals	Bansal, D. K. Miyake, C-C. Chen, S. Groh, R. Williamson, P. L. McNeil, K. P. Campbell "Defective membrane repair in dysferlin-deficient muscular dystrophy" Nature 423: 168-172 , May-2003
Articles in Peer-reviewed Journals	Howdieshell TR, Webb WL, Sathyanarayana, McNeil PL "Inhibition of inducible nitric oxide synthase results in reductions in wound vascular endothelial growth factor expression, granulation tissue formation, and local perfusion." Surgery. 133:528-37 , Oct-2003
Articles in Peer-reviewed Journals	Miyake, K. and McNeil, P.L "Mechanical Injury and Repair of Cells" Critical Care Medicine. 31(8 Suppl): S496-501 , Mar-2003